

REVIEW and CURRENT STATUS (PMAAD COURSE, 2024-25/QII) for PREPROCESSING ANALYSIS			
TASK	Tools/Methods	Current Status	Suggestions
Formatting/Structural Problems	Profiling and Description of Raw Data		
Missing Data	1. Detection and codification of Missing data. 2. Types of Missing Data (MCAR, MAR, MNAR) 3. Imputation 3.1 Basic imputation (mean, median, mode and similar) 3.2 MIMMI 3.3 MICE 3.4 KNN 3.5 Imputation for Time Series		

	3.6 ML approaches (i.e., use of regressors like decision trees, linear regression, random forests) 3.7 Maximum Likelihood Estimation		
Outliers	1. Detection & Diagnosis 2. Outliers treatment		
Feature Engineering - Feature Selection - Feature Extraction - Transformation - Unbalanced databases	1. Feature selection by bivariate analysis, PCA and biplots 2. Instance selection by filter and wrapper methods (instance selection) 3. Feature selection (weighting, filtering, wrapping methods) 4. Feature Extraction – Reviewed 5. Transformation (basics: encoding, normalization and similar) 6. More advanced transformations (BOX-COX for instance) 7. Multivalued variables		

	8. Second generation of knowledge-driven variables and Third generation of data-driven indicators 9. Unbalanced databases		
Visualization	Uses of visualization tools to preprocess data		